02_backtest_stats

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1 02 — Statistical Tests: Diebold-Mariano and the Model Confidence Set

Notebook 01 produced per-origin MAE for each of the ten models at 60 evenly-spaced forecast origins. This notebook asks two rigorous questions of that output:

1. **Diebold-Mariano (pairwise):** is each model significantly different from the RWD baseline in predictive accuracy?
2. **Model Confidence Set (joint):** which models are statistically indistinguishable from the best one, with a built-in correction for multiple comparisons?

These are complementary tools. DM gives familiar pairwise p-values; MCS returns a *set* and handles the multiple-comparisons problem that arises with ten models.

The CSVs written by notebook 01 are loaded directly; no re-running the backtest.

1.1 1. Setup

[7]:
```
import sys
from pathlib import Path

# Make `coffee_forecast` importable whether or not `pip install -e .`
# is in effect for this kernel. No-op when the package is already on
# sys.path (e.g. via the editable install); falls back to inserting
# the repo root so the notebook also runs from a fresh clone.
try:
    import coffee_forecast  # noqa: F401
except ModuleNotFoundError:
    sys.path.insert(0, str(Path.cwd().parent))

import matplotlib.pyplot as plt
from matplotlib.lines import Line2D
import numpy as np
import pandas as pd

from coffee_forecast import diebold_mariano_test, model_confidence_set
from coffee_forecast.config import COLOR_MAP, CSV_DIR, FIG_DIR, SEED, REPO_ROOT

FIG_DIR.mkdir(parents=True, exist_ok=True)
```

1.2 2. Load the 60-origin results

We work exclusively at scale = 60 in this notebook. The smaller scales (1, 10, 30) are useful as robustness checks but dilute the main statistical story.

[8]:
```python
summary = pd.read_csv(CSV_DIR / "summary_all_scales.csv")
results_60 = summary[summary["scale"] == 60].copy()

n_models = results_60["model"].nunique()
n_origins = results_60["origin_id"].nunique()
print(f"Loaded {len(results_60)} rows: {n_models} models × {n_origins} origins.
↪")
```

```
Loaded 600 rows: 10 models × 60 origins.
```

1.3 3. Leaderboard

Mean metrics across the 60 origins, sorted by MAE. This is the descriptive table — the "what happened" summary that every later analysis zooms into.

[9]:
```python
leaderboard = (
    results_60.groupby("model")[["MAE", "RMSE", "MASE", "sMAPE"]]
    .mean()
    .sort_values("MAE")
)

leaderboard.to_csv(CSV_DIR / "final_leaderboard_60.csv")
print("Saved:", (CSV_DIR / "final_leaderboard_60.csv").relative_to(REPO_ROOT))
leaderboard.round(3)
```

```
Saved: results/csv/final_leaderboard_60.csv
```

[9]:

model	MAE	RMSE	MASE	sMAPE
RWD	12.789	14.841	6.103	0.083
ETS	12.819	14.893	6.110	0.085
Theta	12.862	14.934	6.133	0.085
Naive	12.909	14.986	6.147	0.085
RF	13.132	15.188	6.282	0.087
ARIMA	13.239	15.385	6.283	0.085
GARCH	13.505	15.740	6.504	0.090
Granite-TTM	13.842	16.089	6.551	0.092
Ridge	14.682	17.160	6.872	0.101
Prophet	22.687	24.994	10.603	0.152

RWD has the lowest mean MAE, followed by the other simple baselines (ETS, Theta, Naïve) within 0.15 MAE of it. Prophet sits well outside the cluster. The statistical tests below determine which of these gaps are distinguishable from sampling noise.

1.4 4. Diebold-Mariano vs. RWD

Tests the null hypothesis that each model has equal predictive accuracy to the RWD baseline. The loss series is per-origin MAE (one number per forecast origin, 60 total), and HAC variance is estimated across origins using the Bartlett kernel with the Newey-West (1994) rule-of-thumb bandwidth $\lfloor 4(T/100)^{2/9} \rfloor$. The Harvey-Leybourne-Newbold (1997) small-sample correction is applied to reduce over-rejection at T=60: the statistic is multiplied by $\sqrt{(T+1-2h+h(h-1)/T)/T}$ (with $h=1$ this is $\sqrt{(T-1)/T}$) and referenced against the Student-t distribution with $T-1$ degrees of freedom rather than $\mathcal{N}(0,1)$.

- **Positive DM statistic** => the model is *worse* than RWD
- **Negative DM statistic** => the model is *better* than RWD
- Two-sided p-value under DM $\sim t(T-1)$

[10]:
```
# Per-origin MAE dict, ordered by origin_id so the HAC lags line up␣
↪chronologically.
losses = {
    m: results_60[results_60["model"] == m].sort_values("origin_id")["MAE"].
↪values
    for m in results_60["model"].unique()
}

model_order = leaderboard.index.tolist()  # leaderboard order (best → worst)
rwd_mae = leaderboard.loc["RWD", "MAE"]

dm_results = {}
rows = []
for m in model_order:
    if m == "RWD":
        rows.append(
            {
                "Model": "RWD",
                "Mean MAE": rwd_mae,
                "Δ vs RWD": 0.0,
                "DM stat": None,
                "p-value": None,
                "sig": "(baseline)",
            }
        )
        continue
    dm_stat, p_val = diebold_mariano_test(losses[m], losses["RWD"])
    dm_results[m] = (dm_stat, p_val)
    sig = (
        "***"
        if p_val < 0.001
        else "**"
        if p_val < 0.01
        else "*"
```

```
        if p_val < 0.05
        else "†"
        if p_val < 0.10
        else "ns"
    )
    rows.append(
        {
            "Model": m,
            "Mean MAE": leaderboard.loc[m, "MAE"],
            "Δ vs RWD": leaderboard.loc[m, "MAE"] - rwd_mae,
            "DM stat": dm_stat,
            "p-value": p_val,
            "sig": sig,
        }
    )

dm_table = pd.DataFrame(rows).set_index("Model")
print(
    "Significance:  † p<0.10   * p<0.05   ** p<0.01   *** p<0.001   ns = not␣
 ↪significant"
)
dm_table.round(4)
```

```
Significance:  † p<0.10   * p<0.05   ** p<0.01   *** p<0.001   ns = not
significant
```

[10]:

Model	Mean MAE	Δ vs RWD	DM stat	p-value	sig
RWD	12.7889	0.0000	NaN	NaN	(baseline)
ETS	12.8187	0.0298	0.1651	0.8694	ns
Theta	12.8620	0.0731	0.6550	0.5150	ns
Naive	12.9092	0.1203	0.7562	0.4525	ns
RF	13.1320	0.3432	0.5374	0.5930	ns
ARIMA	13.2394	0.4505	1.5235	0.1330	ns
GARCH	13.5048	0.7159	1.7609	0.0834	†
Granite-TTM	13.8416	1.0527	2.3526	0.0220	*
Ridge	14.6819	1.8930	2.0401	0.0458	*
Prophet	22.6871	9.8982	4.5341	0.0000	***

Three patterns in the table:

- **None of the top-tier simple models differ significantly from RWD.** ETS, Theta, and Naive all have $p > 0.4$. The gaps between them fall within sampling noise.
- **Granite-TTM is significantly worse than RWD at the 5% level** (DM = +2.35, $p = 0.022$). Its 32-year pretraining and 1,536-day context window do not improve point accuracy relative to the drift baseline.
- **Prophet is rejected at the 0.1% level** (DM = +4.53, $p < 0.001$), consistent with its mean MAE of 22.7.

Note that DM tests each model against a single baseline (RWD) without multiple-comparisons correction. The MCS in §5 is the proper joint test.

1.5 5. Model Confidence Set

The DM test compares each model to a fixed baseline. The MCS asks the joint question: *which subset of models contains the true best with confidence* $1 - \alpha$?

Procedure (Hansen, Lunde & Nason, 2011):

1. Start with all ten models in the active set.
2. Compute the max-t statistic across centered loss differentials.
3. Bootstrap the null distribution (circular block bootstrap, $B = 5000$, block size 5) to obtain a p-value.
4. If $p \leq \alpha$, eliminate the model with the highest t-statistic and repeat. If not, stop: the active set is the MCS.

Models surviving in the MCS are jointly indistinguishable from the best model at significance level α. Per HLN, p-values reported for eliminated models are monotonically adjusted (the running maximum across elimination steps), so a model eliminated later cannot have a smaller MCS p-value than one eliminated earlier.

Note on power. $T = 60$ origins is on the small side for MCS (HLN's simulations use $T = 100$–500). With $T = 60$ and ten models the test has limited power, so a wider-than-true MCS is partly a power story rather than purely a true-tie story. The qualitative finding below (Prophet eliminated; the other nine retained) is robust, but the surviving-9-model set could plausibly shrink with more origins.

[11]:
```python
print("Running Model Confidence Set (alpha=0.10, block bootstrap B=5000)...")
mcs_models, mcs_df = model_confidence_set(
    losses, alpha=0.10, block_size=5, n_boot=5000, seed=SEED
)

print(f"\nMCS at   = 0.10  ({len(mcs_models)} of {len(losses)} models):")
print("  " + ", ".join(mcs_models))

excluded = [m for m in losses if m not in mcs_models]
print(f"\nExcluded ({len(excluded)}): {', '.join(excluded)}")

mcs_df
```

```
Running Model Confidence Set (alpha=0.10, block bootstrap B=5000)…

MCS at   = 0.10  (9 of 10 models):
  Naive, RWD, Theta, ETS, ARIMA, GARCH, Ridge, RF, Granite-TTM

Excluded (1): Prophet
```

[11]:

	model	mcs_p_value	in_mcs
0	Naive	1.0000	True

```
1          RWD      1.0000    True
2        Theta      1.0000    True
3          ETS      1.0000    True
4        ARIMA      1.0000    True
5        GARCH      1.0000    True
6        Ridge      1.0000    True
7           RF      1.0000    True
8  Granite-TTM      1.0000    True
9      Prophet      0.0088   False
```

1.6 6. Visualization: `dm_mcs.png`

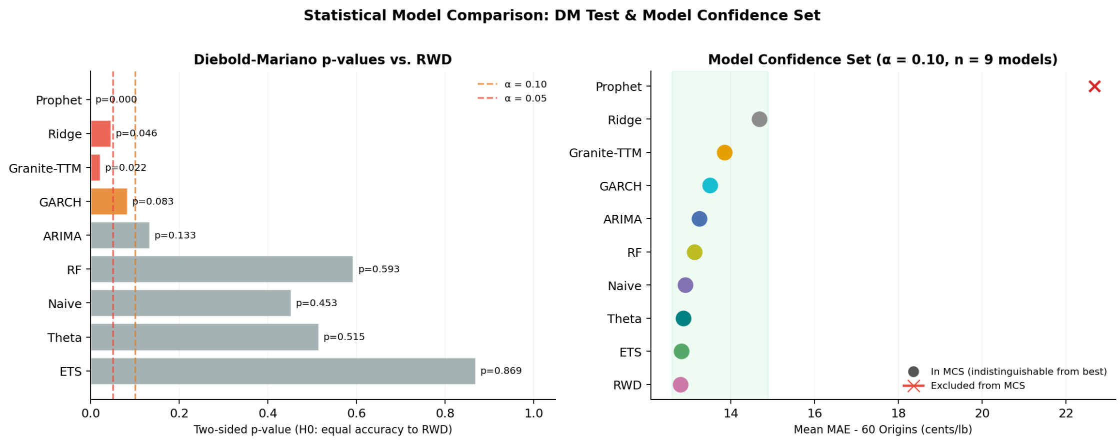
A two-panel figure summarizing both tests.

Left panel: DM p-values vs. RWD, color-coded by significance level. **Right panel:** mean MAE at 60 origins with MCS membership marked — survivors as filled circles, eliminated models as red x's, and the full MCS range shaded green.

[12]:
```
fig, axes = plt.subplots(1, 2, figsize=(13, 5), dpi=180)

# ===== Left: DM p-values vs RWD =====
ax = axes[0]
dm_models = [m for m in model_order if m != "RWD"]
dm_pvals = [dm_results[m][1] for m in dm_models]
bar_colors = [
    "#E74C3C"
    if p < 0.05
    # significant
    else "#E67E22"
    if p < 0.10
    # marginal
    else "#95A5A6"  # not significant
    for p in dm_pvals
]

bars = ax.barh(dm_models, dm_pvals, color=bar_colors, alpha=0.85, edgecolor="white")
ax.axvline(0.10, color="#E67E22", lw=1.4, ls="--", alpha=0.8, label="  = 0.10")
ax.axvline(0.05, color="#E74C3C", lw=1.4, ls="--", alpha=0.8, label="  = 0.05")
for bar, p in zip(bars, dm_pvals):
    ax.text(
        min(p + 0.01, 0.95),
        bar.get_y() + bar.get_height() / 2,
        f"p={p:.3f}",
        va="center",
        fontsize=8,
    )
```

```python
ax.set_title("Diebold-Mariano p-values vs. RWD", fontsize=11, weight="semibold")
ax.set_xlabel("Two-sided p-value (H0: equal accuracy to RWD)", fontsize=9)
ax.set_xlim(0, 1.05)
ax.legend(fontsize=8, frameon=False)
ax.grid(True, axis="x", alpha=0.12)
ax.spines["top"].set_visible(False)
ax.spines["right"].set_visible(False)

# ===== Right: MCS membership dot plot =====
ax = axes[1]
mcs_plot = mcs_df.copy()
mcs_plot["mean_mae"] = [
    leaderboard.loc[m, "MAE"] if m in leaderboard.index else np.nan
    for m in mcs_plot["model"]
]
mcs_plot = mcs_plot.sort_values("mean_mae")

for _, row in mcs_plot.iterrows():
    color = COLOR_MAP.get(row["model"], "#666666")
    marker = "o" if row["in_mcs"] else "x"
    size = 120 if row["in_mcs"] else 80
    zorder = 4 if row["in_mcs"] else 3
    ax.scatter(
        row["mean_mae"],
        row["model"],
        color=color,
        marker=marker,
        s=size,
        linewidths=2.0,
        zorder=zorder,
    )

# Shade the MCS range so the "top tier" is visually obvious.
mcs_maes = mcs_plot[mcs_plot["in_mcs"]]["mean_mae"]
if len(mcs_maes) > 0:
    ax.axvspan(
        mcs_maes.min() - 0.2,
        mcs_maes.max() + 0.2,
        color="#2ECC71",
        alpha=0.08,
        label="MCS range",
    )

legend_elements = [
    Line2D(
        [0],
        [0],
```

```
        marker="o",
        color="w",
        markerfacecolor="#555",
        markersize=10,
        label="In MCS (indistinguishable from best)",
    ),
    Line2D(
        [0],
        [0],
        marker="x",
        color="#E74C3C",
        markersize=10,
        linewidth=2,
        label="Excluded from MCS",
    ),
]
ax.legend(handles=legend_elements, fontsize=8, frameon=False, loc="lower right")
ax.set_title(
    f"Model Confidence Set (  = 0.10, n = {len(mcs_models)} models)",
    fontsize=11,
    weight="semibold",
)
ax.set_xlabel("Mean MAE - 60 Origins (cents/lb)", fontsize=9)
ax.grid(True, axis="x", alpha=0.12)
ax.spines["top"].set_visible(False)
ax.spines["right"].set_visible(False)

fig.suptitle(
    "Statistical Model Comparison: DM Test & Model Confidence Set",
    fontsize=12,
    weight="semibold",
    y=1.01,
)
fig.tight_layout()
fig.savefig(FIG_DIR / "dm_mcs.png", dpi=300, bbox_inches="tight")
plt.show()
print("Saved:", (FIG_DIR / "dm_mcs.png").relative_to(REPO_ROOT))
```



Saved: `results/figures/dm_mcs.png`

1.7 7. Interpretation

DM and MCS look superficially contradictory: DM rejects equal accuracy between RWD and Granite ($p = 0.022$) and Ridge ($p = 0.046$), while MCS keeps all nine non-Prophet models in the confidence set with $p = 1$. The gap is real and explainable. DM tests one pair at a time without correction, so running nine pairwise tests against RWD inflates Type I error; MCS tests the full set jointly and corrects for multiple comparisons by construction. With ten models the joint test is the appropriate primary tool; DM is a familiar pairwise summary that supplements the joint result, not a contradiction of it.

The headline: 9 of 10 models are jointly indistinguishable from the best at $\alpha = 0.10$; only Prophet is eliminated. The nine survivors span every category in the model suite (simple baselines, classical statistical, volatility, recursive ML, and the foundation model), and the differences between them are not statistically resolvable with 60 origins.

So the leaderboard numbers from §3 are descriptive, not a true ranking. RWD isn't "the winner"; it's the arithmetic leader of a cluster of statistically-equivalent models, and there isn't enough signal that this suite can extract to produce one.

Brewing conclusion. A diverse 10-model suite — ranging from random-walk baselines to a foundation model with 32 years of pretraining — cannot statistically distinguish itself from the naive drift forecaster on this series. That pattern has a natural reading in financial economics: it's the empirical fingerprint of *weak-form market efficiency* (Fama 1970), the hypothesis that past prices alone contain no exploitable information about future prices. But this argument has a hole — maybe the suite is the wrong instrument and a different class of models would extract structure that ours misses. Closing that hole requires looking at the series itself, independently of any model. That's what notebook 03 does.

Next notebook: `03_forecastability.ipynb` — asks *why* the benchmark collapses into a statistical tie, using a priori forecastability diagnostics (Spectral Predictability Ω, Permutation Entropy, Hurst exponent + Lo's modified R/S test).